# Lasso and Selective Inference

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## 1 Introduction

### 1.1 Dataset Description

This project uses the **Default of Credit Card Clients** dataset from Taiwan (Yeh & Lien, 2009), which is available on the UCI Machine Learning Repository and Kaggle. The dataset contains 30,000 credit card customers observations ($n$) with a binary outcome, default payment next month, and 23 candidate predictors: demographics (credit limit, sex, education, marital status, age), and six months of repayment history, bill statement amounts, and previous payment amounts (April–September 2005).

The choice of financial credit dataset is because in finance, lenders really care about why a model makes a decision. The standard lasso approach is widely used here because it is easy to understand. In contrast to models like random forests or neural networks, it creates a simple baseline that people can compare. For that reason in this paper we will make a very practical comparison: we will look at the trade-off between a statistically checked model (lasso with selective inference) and an accurate but hard-to-explain model (random forest).

What is important, the monthly data have a strong correlation over time. This correlation structure is perfect for testing selective inference, because selective inference is often used for highly correlated groups of data.

Also, about 22% of the clients in our dataset defaulted, which is a realistic baseline. This is slightly imbalanced, but because the dataset is so large, both the lasso and selective inference should run smoothly and give stable results.

However, class imbalance is more of a problem for the random forest. As Stelzer (2019) pointed out, the way we balance the data can change which variables the random forest thinks are important. To handle this, we will calculate the random forest's variable importance in two ways: using the original, unadjusted data, and using a balanced version of the data.

### 1.2 Literature Review

Selective inference for the lasso was introduced by Lee et al. (2016). They showed that when the lasso selects variables, it changes the math we use to calculate confidence intervals and $p$-values. If we ignore this, our results will be biased (post-selection bias).

Kammer et al. (2022) additionally benchmarked several selective-inference procedures for the lasso (sample splitting, the exact conditional approach of Lee et al). Their tests included logistic models with a 10% to 50% outcome rate, which helps us understand what to expect with our dataset's 22% default rate.

Originally, Yeh & Lien (2009) tested several models on this dataset and found that no single model is always the best. Later, Stelzer (2019) ran a much larger test comparing 23 different machine learning models across four credit datasets. She found that ensemble models, like random forests, usually predict defaults better than simple logistic regressions. And recently, Li et al. (2024) compared a lasso-based model to a random forest. However, they evaluate the lasso's choices at face value, leaving open the question of whether these selected coefficients are statistically valid given that the selection was data-dependent. This paper addresses this limitation by applying the exact post-selection inference methodology of Lee et al. (2016). Instead of comparing raw variable lists, in this study we will isolate the subset of lasso variables that survive and compare this statistically validated set against the random forest's importance rankings.

### 1.3 Research Objectives

Before looking at the actual data, we want to answer this main question: Which variables does the lasso pick as important, which of these stay significant after adjusting for selective inference, and how does this list compare to a random forest's ranking?

We will focus on three smaller questions:

1. Do the lasso and the random forest agree on which variables are the most important for predicting defaults?
2. Since the monthly payment variables are highly correlated, does the lasso pick just one month while the random forest spreads the credit across all months? If so, does selective inference show this uncertainty by giving us wider confidence intervals?
3. Does adjusting for selective inference actually change which variables we label as "significant" compared to a naive analysis that ignores the selection process?

## 2 Data Preparation

### 2.1 Data Integrity Checks and Cleaning

Before feature construction and data partitioning, we verified the structural integrity of the dataset. First, we confirmed that all $n = 30,000$ observations correspond to unique financial clients by verifying the uniqueness of the `ID` identifier variable; no duplicate entries were found. Second, missing value auditing confirmed zero missing data fields across all 23 initial features.

Next, we examined categorical attributes for undocumented level codes. In the raw records, `EDUCATION` contains values coded as 0, 5, and 6 (465 rows, representing 1.55% of the sample). Similarly, `MARRIAGE` contains observations assigned a value of 0 (0.18% of the total sample). Because these undocumented codes represent a minor fraction of the data, listwise deletion could introduce sample selection bias. Consequently, we recoded these values into the designed "other/unknown" classifications (`EDUCATION` $= 4$ and `MARRIAGE` $= 3$).

An examination of the target outcome (`default`) confirmed a baseline default rate of exactly 22.12%. This baseline balance motivates our subsequent evaluation of Random Forest models under both standard and balanced bootstrap sampling schemes (Stelzer, 2019).

Finally, we audited the six monthly bill statement attributes (`BILL_AMT1` through `BILL_AMT6`)

for negative values. Between 1.97% and 2.29% of clients show negative monthly balances. These values were verified to represent valid financial credit balances (such as prepayments or carried-forward refunds) rather than data input errors. We kept them as they were so the credit utilization scores would correctly show the overpayments.

### 2.2 Train/Test Split and Data Leakage Prevention

The cleaned dataset was partitioned into a 70% training sample ($n = 21,000$) and a 30% out-of-sample testing set ($n = 9,000$). We split the data carefully using stratified random sampling so that both the training and test sets had the exact same default rate of 22.12

To prevent data leakage, we did this split before creating any new features. Any rules that depended on overall statistics, like the cutoffs used to divide credit limits into risk tiers, were calculated using only the training data, and those exact same cutoffs were then used on the test set. Features calculated line-by-line for each customer (like payment trends or coverage ratios) naturally avoid data leakage on their own.

## 3 Feature Engineering framework

We built our new features using the six-month history, making sure every feature has a clear financial meaning.

Importantly, we kept the original monthly data right alongside our new summary features. Keeping both lets us test if Lasso needs clean summary scores to work well, or if tree models (in our case Random Forests) can figure out the same patterns on their own from the correlated monthly data.

All of these new features are created automatically using a single function (`engineer_features()`) across four key financial areas:

1. **Demographics (7 features):** We grouped customer ages (`AGE`) into four life stages: young ($< 25$), adult (25–39), midlife (40–59), and senior (60+). We also divided continuous credit limits (`LIMIT_BAL`) into three equal groups: low, medium, and high credit limit tiers. Splitting them into groups helps linear models catch sudden jumps in risk, like how default rates jump from 14.7% for high limits to 30.1% for low limits.

2. **Repayment satus features (Raw 6 + Engineered 18 = 24 Features):** We summarized the raw monthly payment delays (`PAY_0` through `PAY_6`) into overall metrics, including average delay (`avg_delay`), maximum delay (`max_delay`), and delay range (`delay_range`). We also counted total months delayed and months with no activity, along with binary flags identifying clients who always paid on time or were ever seriously delinquent (2 or more months late). To capture trajectory, we fitted linear regressions per client for an overall trend line (`delay_trend`), calculated recent momentum(`delay_momentum`) comparing recent vs. older 3-month averages, and tracked step-by-step month-to-month changes (`delta_delay`). Clients with worsening delays had a much higher default rate (42.2%) compared to those with stable or improving payment histories (16.8%).

3. **Bill amount features (Raw 6 + Engineered 33 = 39 Features):** To construct monthly credit utilization ratios raw bill balances were normalized relative to credit limits ($\text{util}_i = \text{BILL_AMT}_i/\text{LIMIT_BAL}$), applying a lower bound on limits to avoid division-

by-zero errors. For both raw bill amounts and utilization ratios, we extracted identical summary metrics, as we did with repayment status features. We recorded months with negative balances (`n_months_negative_bill`) and flagged clients who ever maxed out their card: `ever_maxed_out` ($\max(\text{util}_i) > 0.9$); clients exceeding this threshold defaulted at a rate of 27.5% as versus 20.2% for non-stressed accounts.

4. **Payment amount features (Raw 6 + Engineered 25 = 31 Features):** Monthly payment amounts (`PAY_AMT1` through `PAY_AMT6`) were converted into monthly coverage metrics ($\text{PAY_AMT}_i / \max(\text{BILL_AMT}_{i+1}, 1)$), to measure how much clients actually paid by comparing monthly payments to the prior statement bill (`pay_to_bill_ratio`). We also created monthly flags showing whether the balance was settled (`paid_in_full`). Borrowers who failed to settle their full balance across all six observed cycles defaulted at a rate of 25.9%, compared to 17.2% for borrowers with at least one full settlement. And continuing the same engineering approach as before, we summarized the 6-month payment history.

In total, the expanded feature matrix contains 105 candidate predictors alongside the account `ID` and binary target variable.

# 4 Preparation for Lasso and Selective Inference

To prepare the dataset for Lasso regularized regression and selective inference, we processed the training and test sets to maintain exact feature alignment.

1. **Categorical encoding:** Categorical attributes (`SEX`, `EDUCATION`, `MARRIAGE`, `age_group`, `limit_tier`) were expanded into full binary dummy matrices via `model.matrix()`. The transformation structure was derived from the training data and applied directly to the test set.

2. **Check for exact linear dependence:** Because engineered summaries (e.g., total sum of bill statements) were included together with raw monthly inputs, linear dependencies occurred within the expanded feature matrix. We checked the training design matrix for redundant columns to identify and eliminate these duplicate linear combinations, ensuring that remaining features provided unique information.

3. **Multicollinearity Diagnostics:** To analyze the underlying dependencies among the raw time-series variables, we computed a correlation matrix across the monthly blocks. As shown in Illustration 0, pairwise correlation analyses revealed strong linear dependencies across adjacent monthly bill statements and payment delay indicators. This high internal correlation serves as an empirical setup for testing Lasso selection and post-selection inference.

4. **Feature Standardization:** All continuous predictors were standardized to zero mean and unit variance. Scaling parameters $(\mu, \sigma)$ were estimated strictly from the training partition and subsequently applied to test set features to ensure uniform $L_1$ penalty weighting.

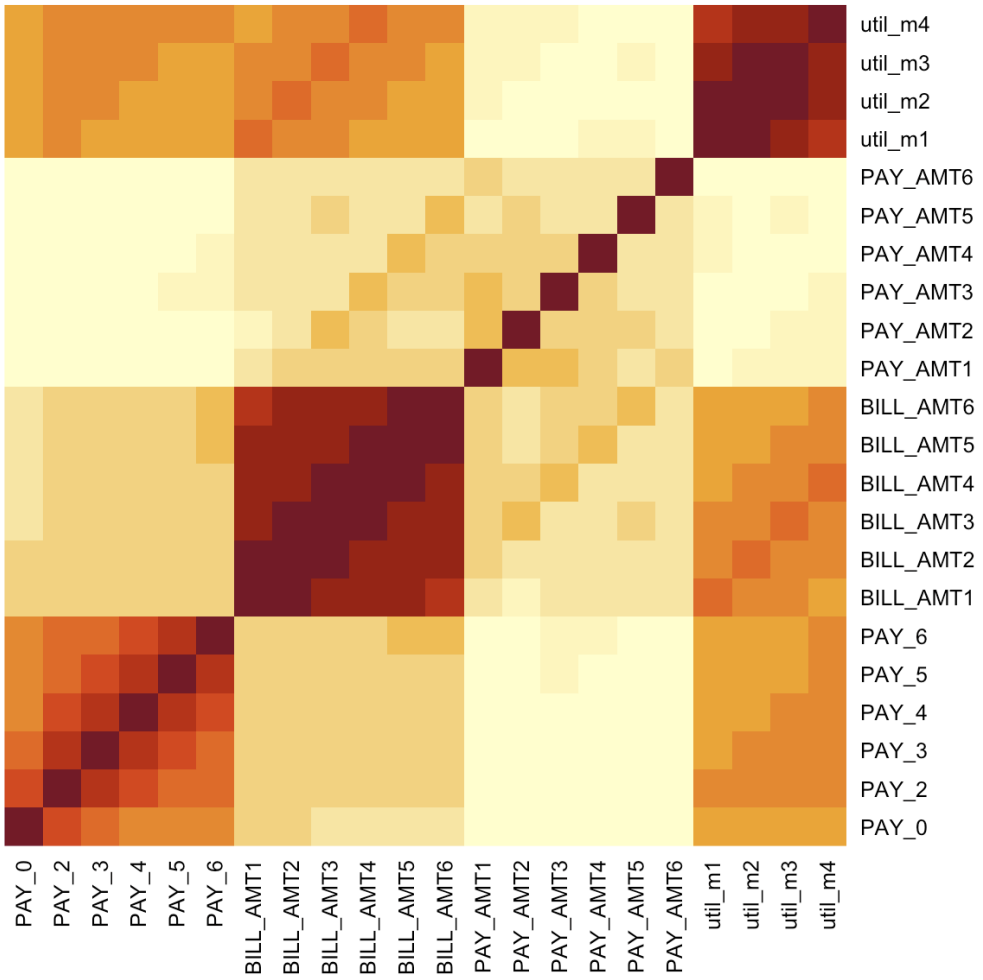



**Illustration 0:** Correlation matrix of monthly time-series predictors highlighting strong pairwise dependencies across adjacent billing cycles.

# 5 LASSO and Selective Inference

## 5.1 Lasso feature selection

Applying default 10-fold cross-validation on the scaled training dataset yielded an optimal penalty value of $\lambda_{\min} = 0.000269$. Then by fitting the regularized logistic regression model at $\lambda_{\min}$ we compress unnecessary predictor coefficients to zero, resulting in a sparse subset of 58 active features (features selected by Lasso).

It is important to note that using cross-validation to select $\lambda$ on the same training data used for post-selection inference means we are using the data twice. While this is standard practice in applied machine learning, traditional selective inference theory requires $\lambda$ to be set in advance before looking at the data. As a result, the adjusted p-values calculated using cross-validated $\lambda$ should be understood as a practical approximation rather than an exact mathematical guarantee (Kammer et al., 2022).

**Results**

The strongest risk factors for default are past payment delays, especially `avg_delay` ($\hat{\beta} = 0.828$) and `ever_seriously_delinquent` (serious late payments) ($\hat{\beta} = 0.573$), and recent late status like `PAY_0` ($\hat{\beta} = 0.179$). On the other hand, consistent financial habits and timely payments reduce default risk, led by months with no card usage: `n_months_no_consumption` ($\hat{\beta} = -0.523$), older payment history, and higher monthly payment amounts. When dealing with highly related monthly data, Lasso picks one main summary feature like `avg_delay` with a large positive impact and uses negative adjustments on single months to balance the model: (`PAY_2` $= -0.505$).

### 5.2 Selective Inference on Lasso-selected variables

Applying standard inference (p-values and confidence intervals) on features chosen by Lasso leads to Type I error because the same data was used for both model selection and inference (post-selection bias).

To correct this selection bias, we applied the polyhedral inference framework using the `fixedLassoInf()` function from the `selectiveInference` R package. Because polyhedral and full debiased inference for binary outcomes (`family = "binomial"`) are restricted to Gaussian settings, we specified `type = "partial"`, which conditions on the selection event while fixing non-target parameters at their Lasso estimates. The resulting output yields valid conditional point estimates, 95% confidence intervals, selection-adjusted $p$-values, and interval widths.

**Results**

After adjusting for selection bias only 24 out of the 58 Lasso-selected variables remain statistically significant ($p < 0.05$). Clear signs of past payment behavior carry the strongest statistical proof, including past late payment statuses (`PAY_2`, $p < 0.001$), months with no card use (`n_months_no_consumption`, $p < 0.001$), and months with zero payments (`n_months_zero_payment`, $p < 0.001$). Higher monthly payment amounts (`PAY_AMT1` to `PAY_AMT6`) also reliably lower default risk ($p < 0.005$).

On the other hand, 34 variables - including closely related statement balances (`BILL_AMT2`, `BILL_AMT3`) and credit usage ratios - lose statistical significance. This shows that while Lasso includes them to improve overall predictions, their individual effects cannot be separated with confidence.

## 6 Selective Inference vs Naive Inference (RQ3)

What if we ignored the fact that selection happened? To evaluate the impact of selection bias, we refit an unpenalized ordinary logistic regression (glm) using only the 58 Lasso-selected variables. This represents the naive approach that ignores the variable selection step.

**Results**

Comparing the naive against the post-selection adjusted results shows a substantial shift in statistical significance. 8 features lost statistical significance after polyhedral correction, including account limit indicators (`limit_tier_mid_limit`), monthly billing metrics (`BILL_AMT6`, `max_bill_amt`, `bill_amt_sd`), payment amounts (`PAY_AMT2`), and repayment flags (`paid_in_full_m4`, `paid_in_full_m5`, `n_months_delayed`). 24 core variables remained statistically significant ($p <$

0.05) under both naive and corrected inference: such as (`n_months_no_consumption`), zero-payment counts (`n_months_zero_payment`), severe delinquency (`ever_seriously_delinquent`), overall average delay (`avg_delay`), payment history metrics (`PAY_0` through `PAY_6`), and direct payment amounts (`PAY_AMT1`, `PAY_AMT3`–`PAY_AMT6`).

To visually examine these shifts in test statistics and interval bounds, we compare naive logistic regression against polyhedral selective inference across visual metrics:

1. **Significance Level Shift (Illustration 1):** Comparing $-\log_{10}(p)$ values shows that post-selection inference drastically magnifies test statistics for core predictors while dampening secondary ones. Primary behavioral drivers (`n_months_no_consumption`, `PAY_2`) jump from moderate naive values ($-\log_{10}(p) \approx 20$) to the maximum display cap (300). Secondary predictors like `ever_seriously_delinquent` drop sharply (from $-\log_{10}(p) \approx 32.8$ to $\approx 5.6$), shifting toward the $\alpha = 0.05$ threshold ($-\log_{10}(0.05) \approx 1.30$).

2. **Confidence Interval Bounds (Illustration 2):** Plotting point estimates and 95% confidence intervals confirms that post-selection intervals (red) are consistently wider than naive bounds (blue). Key variables like `avg_delay` keep similar estimate values, while weaker features like `max_delay` have wider intervals that stretch toward zero, showing that we are less certain about their true effects after selection.

3. **Confidence Interval Width Expansion (Illustration 3):** Mapping naive against corrected interval widths shows that every variable lies above the $y = x$ line ($\text{CI}_{\text{corrected}} > \text{CI}_{\text{naive}}$). While most baseline features cluster near the origin with minor width increases, monthly correlated features (red) and high-variance metrics suffer dramatic, non-linear expansion (reaching widths up to 4.5 and 7.5).

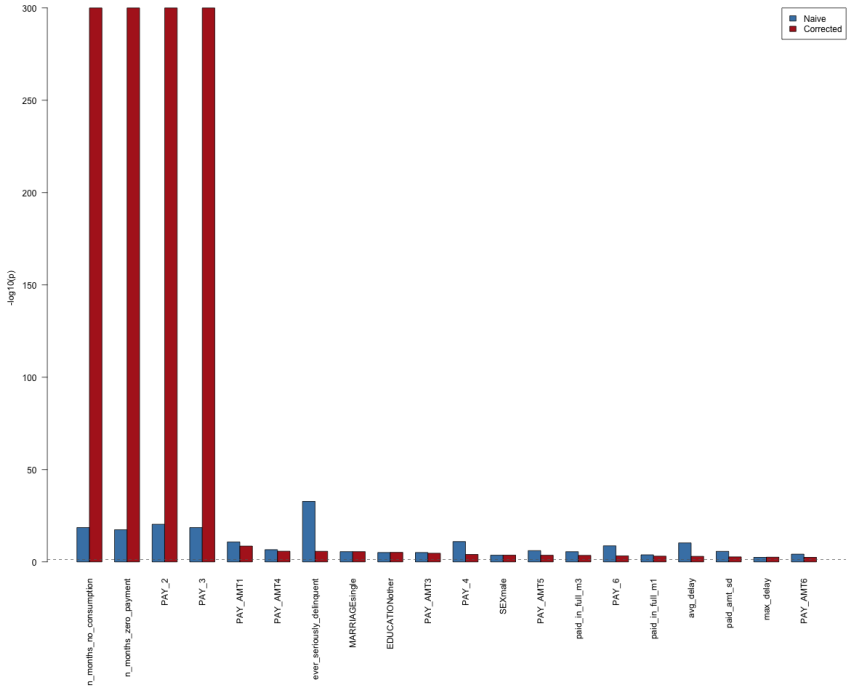



Illustration 1: Naive vs. Corrected $-\log_{10}(p)$-values. Demonstrates the downward shift in statistical significance following conditional selection.

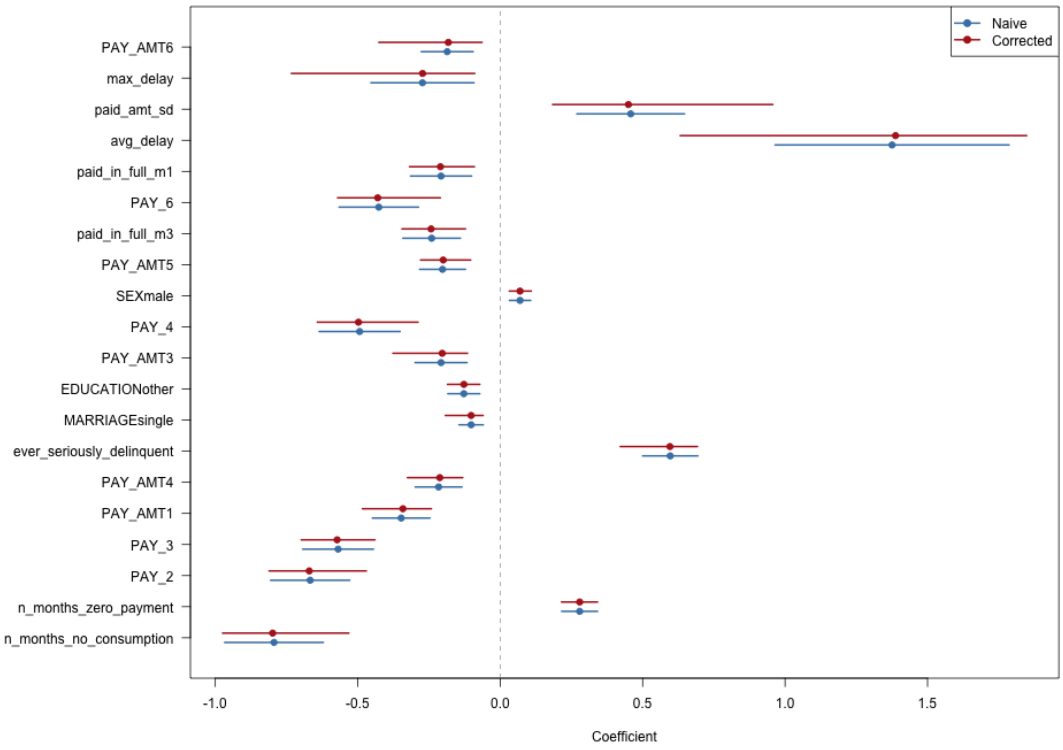



Illustration 2: Forest Plot of Confidence Intervals. Comparing naive GLM point estimates and interval bounds against post-selection selective inference.

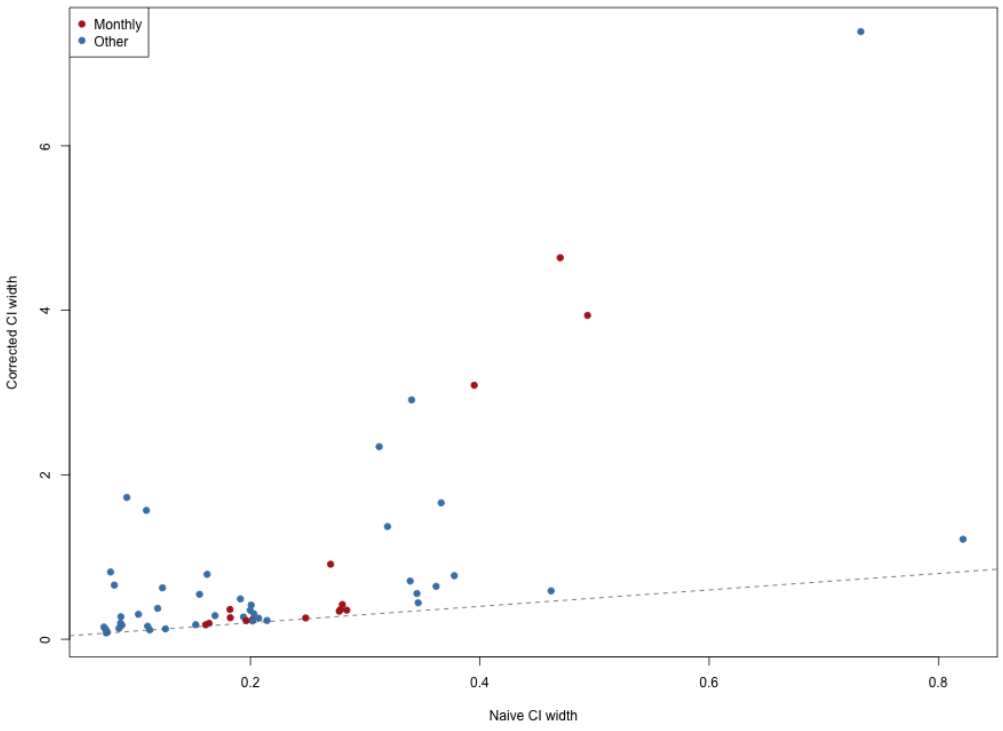



Illustration 3: Scatterplot comparing naive and post-selection confidence interval widths. Points above the equality line reflect interval widening.

# 7 Random Forest Feature Importance under Class Imbalance

To address class imbalance, we trained two distinct Random Forests ($T = 500$ trees each): an unadjusted model (`RF raw`) trained on the natural class distribution, and class-balanced Random Forest (`RF balanced`) using downsampled bootstrap draws (sampsize = $[n_{\text{default}}, n_{\text{default}}]$).

For variable importance we primarly used Mean Decrease Accuracy rather than Mean Decrease Gini. Gini importance measures split purity within individual trees, so it suffers from a well-known bias toward high-cardinality predictors and can be skewed by imbalanced data. Mean Decrease Accuracy directly quantifies the drop in model predictive performance when a variable's values are randomly shuffled - showing how much a feature actually contributes to identifying default risk.

**Results**

The standard model gets a lower total error rate ($0.18 < 0.22$) simply because it guesses the larger group (non-defaulters) most of the time. In contrast, the balanced model gives equal weight to finding both defaulters and non-defaulters. Across all variables, both models generally agree on overall rankings ($\rho = 0.642$). However, when looking at the most important variables, they sharply disagree: only 2 out of the top 10 features overlaps between them.

# 8 Lasso vs. Random Forest Feature Importance Analysis

## 8.1 Rank correlation and top overlaps (RQ1)

To answer to Research Question 1, we compared variable rankings between Lasso and the Class-Balanced Random Forest across continuous predictors, finding a median rank difference of 23.5 across all comparable features.

Two models sharply disagree on primary risk drivers, showing only 1 out of 10 overlapping features in the top 10 (`paid_amt_sd`). Overlap improves only when expanding to the top 30 features (11 out of 30).

Calculating absolute rank differences (rank_diff = $|\text{rank}_{\text{lasso}} - \text{rank}_{\text{rf}}|$) showed that both models agree on raw payment sizes and variations like `paid_amt_sd` (rank_diff $\leq 8$). However, they disagree on risk markers: Lasso favors individual payment statuses such `as PAY _2` and `ever _seriously _delinquent`, while Random Forest prioritizes summary metrics such as credit utilization variability (`util_sd`) and bill amounts.

## 8.2 Correlated Monthly Blocks (RQ2)

To address Research Question 2, we examined model behavior across three six-month temporal blocks: repayment status (`PAY`), bill statement amounts (`BILL_AMT`), and payment amounts (`PAY_AMT`).

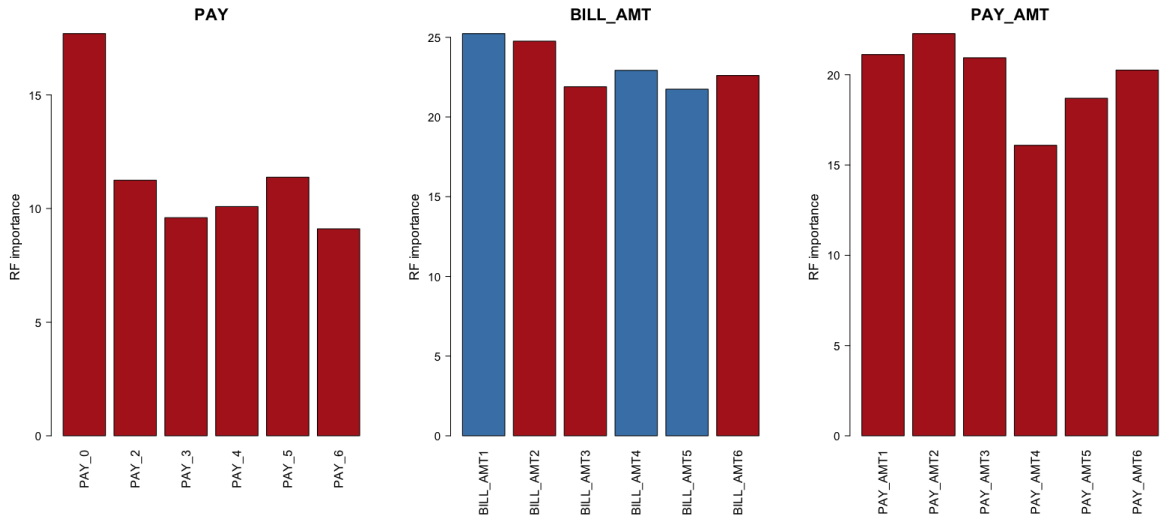



**Illustration 4:** Random Forest variable importance across 6-month temporal sequences, categorized by Lasso selection status (Selected - red vs. Dropped blue).

**Results**

- **W**ithin the `BILL_AMT` block, Lasso retained 3 of the 6 monthly features (dropping Months 1, 4, and 5) to remove redundant linear information. Conversely, Random Forest distributed importance across all 6 months (SD = 1.4697).
- **I**n the `PAY` and `PAY_AMT` sequences, Lasso retained all 6 monthly indicators, showing that each temporal transition provides non-redundant predictive signal. Random Forest placed the highest weight on the most recent month (`PAY_0` importance = 17.69), while maintaining non-zero baseline importance scores across older months.
- **C**orrelated monthly variables suffered from wider post-selection confidence intervals than non-monthly predictors. The mean CI width for selected monthly variables was 1.0517, compared to 0.7767 for all other selected variables. This widening quantifies the estimation uncertainty introduced when Lasso selects among collinear predictors.

# 9 Predictive performance on test set

Finally, we evaluated out-of-sample predictive performance on the held-out test set ($n = 9,000$). Predicted default probabilities were generated for all candidate models to construct Receiver Operating Characteristic (ROC) curves and calculate Area Under the Curve (AUC) metrics.

Table 1: Out-of-Sample Test Set Predictive Performance

| **Model** | **Test-Set AUC** | **95% Confidence Interval** |
|---|---|---|
| Lasso Logistic Regression ($\lambda_{\min}$) | 0.775 | [0.763, 0.787] |
| Unadjusted Random Forest (`RF raw`) | 0.761 | [0.748, 0.773] |
| Class-Balanced Random Forest (`RF balanced`) | 0.773 | [0.761, 0.785] |

The out-of-sample results show that balancing the classes increases Random Forest's AUC from 0.761 to 0.773, bringing its performance almost level with Lasso. But while Class-Balanced Random Forest captures non-linear interactions, regularized Lasso regression performs slightly

better than both Random Forest models. When paired with post-selection inference, Lasso offers a practical balance between predictive performance and statistical interpretability.

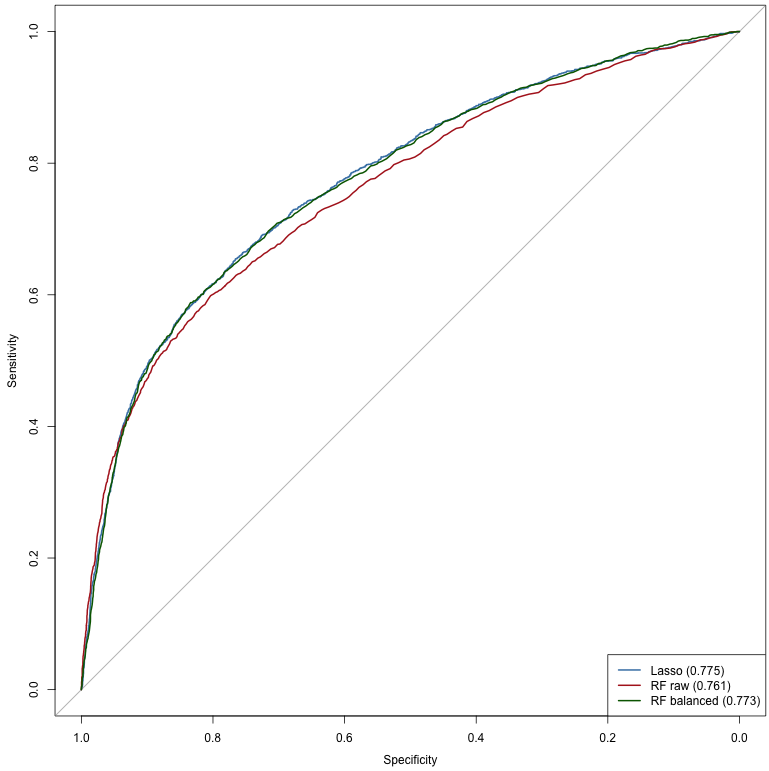



**Illustration 5:** Out-of-sample Receiver Operating Characteristic (ROC) curves comparing Lasso Logistic Regression, Raw Random Forest, and Class-Balanced Random Forest on the test dataset ($n = 9,000$).

## 10 Conclusion of Research Questions

This study provides an empirical evaluation of Lasso feature selection, exact selective inference, and Random Forest variable importance within financial credit risk modeling. Our findings answer the three research questions:

1. **Agreement between Lasso and Random Forest Priority Rankings.** Lasso and Class-Balanced Random Forest display moderate global alignment ($\rho = 0.642$). While global correlation is moderate, the two models disagree on top-tier risk drivers due to structural differences in handling collinear groups and factor variables: Lasso favors individual variables, while Random Forest prioritizes summary metrics.

2. **Model Behavior Across Collinear Temporal Sequences.** Lasso enforces sparsity across highly correlated feature blocks (e.g., bill statements) by selecting subset representatives, whereas Random Forest distributes importance across the entire temporal sequence. Selective inference reflects this selection by generating wider confidence intervals for correlated monthly features.

3. **Practical Impact of Selective Inference Adjustment.** Unadjusted naive logistic regression produces optimistic $p$-values and narrow confidence intervals. Applying exact

polyhedral post-selection inference adjusts for selection bias, widening confidence intervals and filtering out false-positive discoveries.

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